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REVIEW

When physics meets machine learning: a survey of physics-informed machine learning

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Abstract
Physics-informed machine learning (PIML), the combination of prior physics knowledge with data-driven machine learning models, has emerged as an effective means of mitigating a shortage of training data, increasing model generalizability, and ensuring physical plausibility of results. In this paper, we survey a wide variety of recent works in PIML and summarize them from three key aspects: 1) motivations of PIML, 2) physics knowledge in PIML, and 3) methods of physics knowledge integration in PIML. We additionally discuss current challenges and corresponding research opportunities in PIML.

Keywords Machine learning · Data-driven modeling · Physics-informed machine learning · Simulation

1 Introduction

Machine learning/deep learning models have achieved tremendous success in a number of domains, e.g., computer vision [1–5] and natural language processing [6–14], where large amounts of training data and highly expressive neural network architectures together give rise to solutions outperforming previously dominant methods. As a consequence, researchers have also started exploring the possibility of applying machine learning models to advance scientific discovery and further improve traditional analytical modeling [15–21].

Although deep neural networks are able to extract complex relationships between inputs and outputs via optimization over large amounts of data, *prior knowledge* still plays an important role in finding optimal solutions. Prior knowledge, when incorporated properly, can provide rich insight that is otherwise non-existent or difficult to extract from limited training data, and helps improve the data efficiency, generalizability, and plausibility of resulting models.

Physics knowledge, which has been collected and validated explicitly both theoretically and empirically, contains tremendous value due its abstraction of natural phenomena and human behaviour in many important scientific and engineering applications. As a result, in this paper we focus on the topic of integrating prior physics knowledge into machine learning models, i.e., ***physics-informed machine learning (PIML)***. Compared to the integration of other types of prior knowledge, such as knowledge graphs, logic rules, and human feedback [22], the integration of physics knowledge requires particular care due to the many specific properties it captures and analytical forms it can take on.

In this paper, we survey a wide range of recent works in PIML and summarize them from three aspects. (1) Motivations of PIML, which can be further broken into (1a) using machine learning to serve tasks in physics domains and (1b) incorporating physics principles in existing machine learning models for real-world tasks. (2) Physics knowledge in PIML, each class of which corresponds to a general principle with a wide range of applications. (3) Methods of physics knowledge integration in PIML methods, which we further categorize into means of data enhancement (e.g., augmenting training data with physically-grounded synthetic data),

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neural network architecture design, and physics-informed optimization (e.g., physics-augmented loss functions).

The paper is organized as follows. Section 2 analyzes two main categories of motivations using PIML: one mainly serves tasks in physics domains, while the other serves real-world problems. Section 3 introduces several general physics principles widely used in PIML. Section 4 investigates methods of physics knowledge integration. Section 5 discusses challenges and potential future research directions of PIML.

2 Motivations of PIML

2.1 ML for Physics: enhancement of physics models via data-driven methods

Physical science problems involve various data-intensive tasks including spatiotemporal data modeling, causal reasoning, computer vision, probabilistic inference and so on. Since machine learning methods have achieved great success in these tasks, using machine learning models for furthering scientific discovery in physics has received increasing interest in recent years. Compared to existing numerical or pure physics based methods, physics-informed machine learning methods have advantages in flexibility, generalizability, and computational cost while helping to enforce physical plausibility. In this section, we introduce recent developments in leveraging machine learning for several physics-related tasks, including surrogate simulation, data-driven PDE solvers, parameterization of physics models, reduced-order models, and knowledge discovery.

2.1.1 Simulation

Many physics-informed machine learning models have been proposed to act as surrogate solutions to numerical simulation across a variety of domains, such as turbulence simulation [23–25], climate simulation [26–29], and particle system simulation [30–33]. Compared to numerical simulation, neural network based machine learning models enjoy the following advantages: (1) **Lower implementation costs.** To build a high-quality numerical simulator, researchers usually need years of engineering effort and must choose numerous physically meaningful parameters depending on the task. Machine learning models can instead be trained directly from a large amount of observed data. (2) **Stronger ability to generalize.** Machine learning models can share the same architecture for different types of problems in the same category, and then can be further specialized for each problem with observed data. For example, [30] proposes a general framework to learn particle simulation, and can generalize across fluid, rigid, and deformable material systems. (3) **Lower computation costs.** Existing neural network building blocks such as multilayer perceptrons and convolutions can be efficiently accelerated by various hardware (e.g., CPUs, GPUs, FPGAs, ASICs) and yield significant advantages in computational costs for neural networks composed of these blocks. In fluid flow prediction, [34] proposes a model that achieves prediction results close to ground truth simulation at a running speed 60x faster than the numerical method.

Meanwhile, due to the large and complex optimization space of neural networks, it is critical to incorporate inductive bias of physics knowledge about the task into either the training data, the model architecture, or the optimization process. This ensures greater physical plausibility of resulting models, further improving the robustness in real-world settings. The detailed techniques of integration are introduced in Section 4.

2.1.2 Data-driven PDE solver

Many real-world problems from physical systems principally aim to describe observations based on partial differential equations and solve them numerically. There are many traditional methods to numerically solve such PDEs, including spectral methods, finite-difference methods (FDM), finite element method (FEM), finite volume method (FVM), etc. All of these methods are numerical and require proper discretization or a finite number of steps to produce approximate continuous solutions.

PIML methods generally pose a compelling alternative to traditional solvers in practice, although one's confidence in the underlying physics is of the utmost importance when deciding between the two paradigms. When the task in question requires precise outputs and is well-formed with known governing equations, a traditional solver is likely preferable; they often provide known error bounds and are more computationally efficient than a less direct PIML approach. PIML methods generally trade some of this reliability for flexibility, and are thus particularly useful when the prior physics knowledge is not exact. For instance, in many real-world settings, available PDEs may only partially capture the dynamics of the underlying process, in which case models like physics-informed neural networks (PINNs) [35] stand to learn a useful hybrid representation, leveraging both a data-driven and physics-based component. PINNs have been shown to effectively account for missing model parameters, unknown boundary conditions, and generally handle noisy/sparse measurements better than traditional solvers. Additionally, PINN methods can tackle much higher dimensional problems and are mesh-free, allowing for greater reuse once trained due to fast inference across the spatial domain.

Combining machine learning techniques with PDE models more broadly has long received attention from the ML community. Crutchfield and McNamara [36] introduce a method to reconstruct the deterministic portion of the equa-

tions of motion directly from a data series. This approach employs an informational measure of model optimality to guide searching through the space of dynamical systems. Kevrekidis et al. [37] present a framework for computer-aided multi scale analysis, which enables models at a fine (microscopic/stochastic) level of description to perform modeling tasks at a coarse (macroscopic/systems) level. These macroscopic modeling tasks, yielding information over long time and large scales, are accomplished through approximately initialized calls to the microscopic similar for only short times and small spatial domains.

More recently, [38, 39] introduce a solution to solve PDE in a data-driven manner. Rather than analytically solving a given equation, they infer solutions to targeted PDE via supervised learning. Given an input tuple (x, t), they compute spatial and time derivatives from black-box models, after which outputs are connected based on a form of PDE to update all learnable parameters in the black-box models. This method does not require any discretization and is fully data-driven in its construction of a surrogate model.

Raissi [40] approximate the unknown solution as well as the nonlinear dynamics by two deep neural networks. The first network acts as a prior on the unknown solution and essentially avoids numerical differentiations which are inherently ill-conditioned and unstable. The second network represents the nonlinear dynamics and helps distill the mechanisms that govern the evolution of a given spatiotemporal dataset. [41] introduce a technique based on the singular vector canonical correlation analysis (SVCCA) for measuring the generality of neural network layers across a continuously-parameterized set of tasks. They illustrate this method by studying generality in neural networks trained to solve parameterized boundary value problems based on the Poisson partial differential equation. [42] directly model the mapping between a PDE's solution and its initial conditions via message passing in the spatial domain. The following work [43] further extends the modeling to the frequency domain, which inherently generalizes to multiple spatial resolutions.

In many physical systems the governing equations are known with high confidence, but direct numerical solutions are prohibitively expensive to compute. Often this situation is alleviated by writing effective equations to approximate dynamics below the grid scale. This process is often impossible to perform analytically and is mostly ad hoc. Bar-Sinai et al. [44] propose data-driven discretization, a method that uses machine learning to systematically derive discretizations for continuous physical systems. [45] target the problem of reducing numerical errors of iterative PDE solvers and compare different learning approaches for finding complex correction functions. They integrate the PDE solver into the training loop and thereby allow the model to interact with the PDE during training.

Downscaling

Directly solving PDEs requires spatial and temporal discretization, and finer resolutions are often preferred to capture physically reliable solutions. Doing so, however, increases the computational cost and modeling complexity significantly in most cases. To address this, downscaling techniques have been widely used as a solution to capture physical variables that need to be modeled at a finer resolution from a coarser resolution. Recently, artificial neural networks have shown great promise in this domain given their ability to model non-linear relationships. [46] present a downscaling algorithm using neural networks to leverage the relationships between satellite precipitation estimates (SPEs) and microphysical cloud properties in northeast Austria. [47] present DeepSD (Statistically Downscaling), a generalized stacked super-resolution convolutional neural network (SRCNN) framework for statistical downscaling of climate variables. DeepSD augments SRCNN with multi-scale input channels to maximize predictability in statistical downscaling. [48] introduce a data-driven framework for the identification of unavailable coarse-scale PDEs from microscopic observations via machine learning algorithms. Specifically, using Gaussian processes, artificial neural networks, and/or diffusion maps, the proposed framework uncovers the relation between the relevant macroscopic space fields and their time evolution (the right-hand side of the explicitly unavailable macroscopic PDE).

2.1.3 Parameterization

When using numerical models that describe complex physics phenomena, *parameterization* is a technique commonly employed to replace difficult-to-model systems with simpler parameterized processes. Traditionally, parameters are chosen to minimize the discrepancy between model outputs and observed data via common techniques like grid search or Bayesian inference over the parameter space. To subvert the computational costs and improve the accuracy of these methods, recent works have leveraged advances in machine learning to directly approximate unknown dynamics.

In geology, [49] study the application of Wasserstein GAN [50] for the parametrization of geological models. The effectiveness of the method is assessed for uncertainty propagation tasks using several test cases involving different permeability patterns and subsurface flow problems. [51] develop a new predictor for near-bed suspended sediment reference concentration under unbroken waves using genetic programming, a machine learning technique.

In meteorological science, [52] show that a neural network-based parameterization can be successfully trained using a near-global aqua-planet simulation with a 4-km reso-

lution (NG-Aqua). The neural network predicts the apparent sources of heat and moisture averaged onto (160 km^2) grid boxes. [53] present a novel approach to convective parameterization based on machine learning, using an aquaplanet with prescribed sea surface temperatures as a proof of concept. A deep neural network is trained with a super-parameterized version of a climate model in which convection is resolved by thousands of embedded 2-D cloud resolving models.

In chemistry, supervised learning on molecules has incredible potential to be useful in chemistry, drug discovery, and materials science. Gilmer et al.[54] reformulate existing models into a single common framework called Message Passing Neural Networks, and explore additional novel variations within this framework. The authors further use MPNNs to demonstrate state of the art results on an important molecular property prediction benchmark. Wang et al. [55] utilize molecular graph data for property prediction based on spatial graph convolution neural networks. Mortazavi [56] comprehensively explore recent advances in PIML methods, namely machine learning interatomic potentials, for design and discovery of materials for energy systems.

2.1.4 Reduced-order models

Instead of directly modeling complex dynamics in the observation space, researchers have developed simplified, interpretable models (reduced-order models, ROMs) in the hidden space, which can usually be derived from the observation space via dimensionality reduction. In recent works, machine learning models have emerged as effective tools to discover these useful hidden spaces along with information-preserving transformations between them and the observation space.

In dynamic system analysis, Koopman theory is a commonly used ROM. Koopman theory is based on the insight that the state space of a non-linear dynamic system can be encoded into an infinite-dimensional space where the dynamics are linear [57]. In practice, it is often assumed the infinite-dimensional space can be approximated with a finite-dimensional space. The key problem is then to find a proper encoder/decoder pair to map from/to the state space to/from the hidden space. Traditionally, encoder/decoders were constructed with hand-crafted functions, such as the identity function in Dynamic Mode Decomposition (DMD) [58], nonlinear functions in Extended DMD (EDMD) [59], and kernel functions in Kernel DMD (KDMD) [60]. These functions, however, may fail to fit sufficiently complex dynamical systems and are difficult to design without domain-specific knowledge. As a result, recent works [61–63] construct encoders/decoders using neural networks as trainable universal approximators. They demonstrate that the combination of neural networks and Koopman theory achieves comparable or even higher performance than the Koopman approximators with hand-crafted mapping functions, while enjoying the ability to generalize to multiple datasets with the same design. [61] further shows that the integration of Koopman theory allows the model to adapt to new systems with unknown dynamics faster than pure neural networks.

In fluid dynamics, ROMs are largely considered due to their unprecedented physical insight into turbulence offered by high-fidelity computational fluid dynamics (CFD). [64] develop a dimensionality reduction method called Non-Intrusive Reduced Order Model (NIROM) for predicting the turbulent air flows found within an urban environment. Mohan and Gaitonde [65] demonstrate a deep learning based approach to build a ROM using the proper orthogonal decomposition (POD) basis of canonical direct numerical simulation (DNS) datasets, employed for turbulent flow control applications. They find that Long Short Term Memory (LSTM) networks, a type of recurrent neural network that is commonly employed for problems like speech modeling and language translation, shows attractive potential in modeling temporal dynamics of turbulence.

2.1.5 Causality

Causal Discovery and Causal Inference in Time Series

Uncovering the underlying causal structure in systems is a fundamental problem that has received a great deal of attention by many scientific communities over the years. [66–68] introduce the problem and provide a mathematical framework for causal reasoning and inference under causal graphical models (also known as Bayesian networks (BN) [69]). [70] formalize a concept of quantifiable causality in time series, called Granger causality. From the pioneering works, learning causal associations from time series data has emerged as a popular topic in the machine/deep learning community. [71] propose a method to distinguish direct from indirect dependencies and common drivers among multiple time series to reconstruct a causal network. [72] quantify causal associations in nonlinear time series and [73, 74] provide promising applications of the causal discovery in time series. [75] introduce a smooth acyclicity constraint to multivariate time series inspired by [76] that consider causal discovery as a purely continuous optimization problem. [77] further cast light on the role of causality in the spatiotemporal setting, constructing a structural causal model [78] to better represent temporally invariant features and dynamic spatial causation.

For causal inference, G-computation formula, g-estimation of structural nested mean models [79], and inverse probability of treatment weighting in marginal structural models (MSMs), [80, 81] rely on linear predictors for estimation to estimate treatment effects. Recurrent marginal structural networks (RMSNs) [82] are proposed to further improve MSM's

ability by capturing non-linear dependencies. In addition, Gaussian processes [83, 84] have been tailored to estimate treatment responses in continuous-time settings in order to incorporate non-deterministic quantification. Further, [85] and [86] theoretically prove that observed proxy variables can be used to capture hidden confounders and estimate treatment effects. The time series deconfounder (TSD) [87] introduces recurrent neural networks in the factor model to estimate the dynamics of confounders. In a similar vein, [88] propose a sequential deconfounder to infer hidden confounders by using Gaussian process latent variable models. The deconfounding temporal autoencoder (DTA) [89] estimates treatment effects under dynamic setting using observed data as noisy proxies. Deep Sequential Weighting (DSW) [90] infers the hidden confounders by using a deep recursive weighted neural network that combines current treatment assignment and historical information. The Dynamic Networked Observational Data De-confounder (DNDC) [91] aims to learn how hidden confounders behave over time by using current network observational data and historical information. Although many works have demonstrated success in discovering unknown causal structure from observational data, few works leverage explicit causal relations from physical knowledge to improve data-driven models.

Counterfactual Analysis of Physical Dynamics

Counterfactual analysis in the physical world is typically concerned with analytically predicting the effects of various types of interventions/treatments, including the physical laws of the current environment [92–94], the actions of the agent itself [95–98], and the decision outcomes of other agents in multi-agent systems [99, 100]. With the development of 3D simulation in the field of machine learning, several benchmarks performing counterfactual analysis in physics have emerged. CLEVRER [101] asks agents to answer a counterfactual question after observing a video showing the movement and collision of a 3D object. Physics 101 [102] presents a video benchmark containing over 101 real-world objects physically interacting in four different physical scenarios. [103] proposes a counterfactual benchmark with two tasks: a scene where balls interact according to unknown interaction laws (e.g., gravity or elasticity), and a scene where clothes are folded by the wind. The agent needs to discover causal relationships between counterfactual variables and objects and then predicts future frames.

The counterfactual analysis of the physics-informed models relies on the separation between the physical information features and the remaining interactions [104, 105], which may incorporate additional information based on the available prior knowledge of the scene [106]. PhyDNet [107] explicitly disentangles PDE dynamics from unknown complementary information. The interpretable, intuitive physical model [108] proposes an encoder-decoder framework for predicting future collision event frames. The encoder layer infers physical properties, such as mass and friction, from the input frames. Then, the decoder decomposes the potential physical vectors by outputting optical flow. Physical interaction prediction (PIP) [109] uses a deep generative model to build approximate mental simulations by generating a framework for future physical interactions and then employs selective temporal attention in the form of spanwise selection to predict the outcome of physical interactions. Causal World Models (CWMs) [110] allow unsupervised modeling of relationships between the intervened observations and the alternative futures by learning an estimator of the latent confounding factors. Cophy [111] predicts alternative outcomes of a physical experiment by estimating the potential performance of confounding factors. Filtered-Cophy [112] further learns and acts on a suitable hybrid potential representation based on a combination of dense features, sets of 2D keypoints, and an additional latent vector per keypoint.

2.2 Physics for ML: improvement of data-driven models from external knowledge

Data-driven methods represented by machine learning and neural networks have achieved great success recently in a wide range of real-world problems due to the universal approximation ability of neural networks [113], the increase of available data for model training, and the rapid development of hardware neural network accelerators. However, the optimization of neural networks is highly non-convex, and its convergence to global minima is hard to achieve in practice. Optimization processes converging to local minima without constraints may lead to models with limited generalization ability or results violating existing knowledge including commonsense, logic rules, and physics laws [22]. In this section, we focus on the integration of physics knowledge multiple domains of machine learning.

2.2.1 Supervised learning

Typical supervised learning tasks such as classification and forecasting cover various real-world applications from multi-agent systems, computer vision, time series analysis, and spatio-temporal data modeling. While deep neural network-based methods have been the dominating solution in these areas recently, people have also explored several novel approaches with the incorporation of physics knowledge in tasks on data from physics-related scenarios, such as object-centric data, spatio-temporal data, and geometry data.

Object-centric data

Object-centric data is generated by systems composed of multiple discrete objects. Examples of real-world object-centric data include trajectory data from multi-agent sys-

tems [114–116], position and velocity data from networks of motion sensors [115, 117], and molecule data [54]. Graph neural network (GNN)-based methods [92, 116, 118, 119] achieve state-of-the-art results on most tasks with object-centric data due to the matching of its inductive bias [118] and the interacting physics property in object-centric data. [120] demonstrates that GNNs generalize well in many popular object-centric tasks because the forwarding process of GNNs aligns to the underlying reasoning process. In addition to this alignment, other types of physics knowledge are also integrated in recent works. [121] obtains a joint model for the total energy and interatomic forces of molecules that follows the energy-conserving law. [122] proposes Deep Lagrangian Networks (DeLaN) for robotics, on which Lagrangian Mechanics is imposed and physical plausibility is maintained. [114] designs models for forecasting n-body systems that learn and respect exact conservation laws - Hamiltonian mechanics - in an unsupervised manner. Lagrangian Neural Networks (LNNs) in [123] further parameterize arbitrary Lagrangians using neural networks without canonical coordinates or restrictions of the functional form of learned energies, and can be applied to graphs and continuous systems. [61] learns compositional Koopman operators using GNNs and shows better efficiency and generalization than existing GNN baselines in multi-object systems such as ropes and soft robots.

Spatio-Temporal Data

Spatio-temporal data records the dynamics of values of interest in multiple locations within a period of time. The most common form of spatio-temporal data is video, where all locations are aligned to rectangular mesh grids in some specific area. Spatio-temporal data can also be generated from networks of irregularly spaced sensors in domains such as traffic [124], weather [125], and electricity [126]. Since the underlying processes of spatio-temporal data are usually governed by physics laws, physics-informed methods have potentials to further improve the performance of neural network models. [107] explicitly disentangles known PDE dynamics from unknown factors and performs PDE-constrained prediction in latent-space, both of which contribute to video forecasting performance. [127] mimics the pipeline of physical flow simulation, and evolves and accumulates point features in point clouds using flow velocities generated from a high-dimensional force field. It demonstrates the efficacy of the proposed method in various point cloud classification and segmentation problems. [128] learns finite differences of sparsely available data points inspired by physics equations, and shows the superiority in synthetic graph signal prediction and real-world weather forecasting tasks. [129] proposes a general continuous-time differential model for dynamical systems, which admits arbitrary space and time discretizations, and enables efficient neural PDE inference. In the space of traffic forecasting, [130, 131] explicitly incorporate graph representations in neural ODE contexts, better coupling the complex spatial and temporal dependencies when predicting traffic flows. [132] additionally assume traffic flows are generated according to a latent potential energy field, and propose a so-called "physics-guided" model that facilitates an effective trade-off between purely data-driven and physics-based counterparts. [133] propose an active interaction scheme to collect samples from physics-based traffic simulations and efficiently calibrate system parameters, yielding road network models that can be used for forecasting.

Manifold Data

Manifold data describes signals defined on non-planar surfaces such as spheres and surfaces of complex 3D objects, where the Euclidean geometry only holds locally near each point. Examples are magnetoencephalography (MEG) brain activity signal data [134] (on sphere), omnidirectional image data [135] (on 3D objects), and human scan data [136] (on 3D objects). Without the restriction that data points are distributed on planes, general manifold data comes with richer information about the space including local structures and symmetries of the space. Meanwhile, existing convolution-based neural network architectures are either no longer applicable in non-planar manifolds (CNNs) or not capable of fully exploiting the spatial information (GNNs). Instead, physics-informed neural networks incorporate established knowledge of manifolds into the construction of new types of convolutions for manifolds and bridge the gap. [137] constructs intrinsic CNN-like architectures on non-planar surfaces under the geodesic polar coordinate system. [138] alternatively uses the principal curvature directions as fixed gauges to construct convolutions in corresponding tangent fields. [139] constructs convolution kernels as rotating filters and collects the strongest responses among all possible directions. [140, 141] further extends convolution filters to be gauge-equivariant via parallel transporting geometric features to the same vector space before applying the filters.

2.2.2 Model-based control

Due to the integration of physics knowledge, physics-informed machine learning models enjoy better physical plausibility, higher data efficiency and stronger generalization ability compared to pure neural network models, all of which are critical properties for constructing a good model describing the relation between the control input and the state transition of dynamic systems. As a result, PIML methods

also often facilitate greater interpretability than their purely data-driven counterparts. For instance, we generally expect to understand failure modes better in PIML methods: when a model underperforms or fails to generalize, we can usually determine whether this relates to issues with our domain-specific priors (i.e., a bad physics fit) by inspecting model reliance on the provided physical constraints. In many of the architectures discussed above, hidden layers or other intermediate representations can themselves be physically meaningful, in which case their structure can be empirically verified, analyzed, or used for other downstream tasks.

Recent works have shown that physics-informed machine learning models achieve significant success in model predictive control and model-based reinforcement learning. [142] integrates a deep neural network (DNN) that learns high-order interactions in the dynamics model, and constrains the Lipschitz constant of the DNN to guarantee system stability. [122, 143] enforce Lagrangian and Hamiltonian dynamics in the modeling of underlying system dynamics, respectively, and both outperform model learning approaches without physics knowledge in trajectory tracking error, learning speed, and robustness. [61] shows that dynamics learned via compositional Koopman operators can quickly adapt to new environments of unknown physical parameters in online learning. [144, 145] augment partial differential equations (PDEs) that approximately describe continuous physical systems with controllable force terms, and demonstrate that proposed methods successfully control the evolution of complex physical systems.

3 Physics knowledge in PIML

In this section, we introduce several categories of general physics knowledge integrated in PIML. While there is much more domain/task-specific knowledge that can be incorporated for corresponding solutions, each category we introduce in this section covers a wide range of problems and inspires a series of works generally applicable to them instead of leading to only one or two task-specific solutions.

3.1 Classical mechanics and energy conservation laws

Newtonian, Lagrangian, and Hamiltonian mechanics are three typical approaches describing systems of classical mechanics. While Newtonian mechanics has been widely used to describe the relations among locations, velocities, accelerations and forces, Lagrangian and Hamiltonian mechanics provide effective tools to enforce laws of conservation of energy in the modeling of dynamic systems.

Lagrangian Mechanics

Lagrangian mechanics principally define the Lagrangian function L of generalized coordinates $\boldsymbol{q}$ and its gradient w.r.t time $\dot{\boldsymbol{q}}$ to fully describe dynamics of a mechanical system. L is commonly chosen as the difference between the kinetic energy T and the potential energy V, i.e. $L(\boldsymbol{q}, \dot{\boldsymbol{q}}) \equiv T(\boldsymbol{q}, \dot{\boldsymbol{q}}) - V(\boldsymbol{q})$.

The defined Lagrangian function must satisfy the principle of stationary action: the real physical trajectory of a system will always be one where the action is stationary, which is mathematically expressed as:

$$\delta \int_{t_1}^{t_2} L dt = 0. \tag{1}$$

We can further derive the important Euler-Lagrange equation from Eq. 1:

$$\frac{d}{dt} \nabla_{\dot{\boldsymbol{q}}} L = \nabla_{\boldsymbol{q}} L. \tag{2}$$

According to Noether's theorem, the energy of a system is conserved if the system has a time-translation symmetry, i.e., if the Lagrangian function does not explicitly depend on time ($\frac{\partial L}{\partial t} = 0$). Therefore, as long as we model the Lagrangian function as a function of $(\boldsymbol{q}, \dot{\boldsymbol{q}})$ without the explicit time term t and it satisfies Eq. 2, it automatically satisfies the law of conservation of energy. With Eq. 2 we can derive the expression of $\ddot{\boldsymbol{q}}$ with the chain rule [123]:

$$(\nabla_{\dot{\boldsymbol{q}}} \nabla_{\dot{\boldsymbol{q}}}^T L)\ddot{\boldsymbol{q}} + (\nabla_{\boldsymbol{q}} \nabla_{\dot{\boldsymbol{q}}}^T L)\dot{\boldsymbol{q}} = \nabla_{\boldsymbol{q}} L. \tag{3}$$

When L is modeled as a differential function such as that which can be expressed with a neural network, we can solve $\ddot{\boldsymbol{q}}$ as:

$$\ddot{\boldsymbol{q}} = (\nabla_{\dot{\boldsymbol{q}}} \nabla_{\dot{\boldsymbol{q}}}^T L)^{-1} [\nabla_{\boldsymbol{q}} L - (\nabla_{\boldsymbol{q}} \nabla_{\dot{\boldsymbol{q}}}^T L)\dot{\boldsymbol{q}}]. \tag{4}$$

Usually one system's $\boldsymbol{q}$ and $\dot{\boldsymbol{q}}$ can be observed as spatial coordinates and speeds of objects, and the system can be solved with $(\boldsymbol{q}, \dot{\boldsymbol{q}})|_{t=T} = \int_0^T (\dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}) dt$.

For systems with non-conservative forces, Eq. 2 can be extended as

$$\frac{d}{dt} \nabla_{\dot{\boldsymbol{q}}} L - \nabla_{\boldsymbol{q}} L = \boldsymbol{\tau}, \tag{5}$$

where $\boldsymbol{\tau}$ are generalized forces and can be used for controllers such as a PD-Controller [122].

Hamiltonian Mechanics

Hamiltonian mechanics defines the Hamiltonian function H

as a function of a pair of variables $(\boldsymbol{q}, \boldsymbol{p})$, where $\boldsymbol{q}$ is the vector of spatial coordinates of system objects and $\boldsymbol{p}$ is the vector of their momentum. $(\boldsymbol{q}, \boldsymbol{p})$ must satisfy the canonical condition:

$$\boldsymbol{p} = \nabla_{\dot{\boldsymbol{q}}} L, \tag{6}$$

where L is the Lagrangian function of the system. The Hamiltonian function is defined as follows:

$$H(\boldsymbol{q}, \boldsymbol{p}) = \dot{\boldsymbol{q}} \cdot \boldsymbol{p} - L. \tag{7}$$

Combining Eqs. 6, 7, and 2 we can derive:

$$\dot{\boldsymbol{p}} = -\nabla_{\boldsymbol{q}} H, \ \dot{\boldsymbol{q}} = \nabla_{\boldsymbol{p}} H. \tag{8}$$

With Eq. 8, we can verify that

$$\frac{dH}{dt} = \dot{\boldsymbol{q}} \nabla_{\boldsymbol{q}} H + \dot{\boldsymbol{p}} \nabla_{\boldsymbol{p}} H = 0. \tag{9}$$

In the context of classical mechanics, we have $\boldsymbol{p} = M\dot{\boldsymbol{q}}$, $T = \frac{1}{2}\dot{\boldsymbol{q}}^T M \dot{\boldsymbol{q}}$, $L = T - V$, which can then be rewritten with Eq. 7 as $H(\boldsymbol{q}, \boldsymbol{p}) = \dot{\boldsymbol{q}}^T M \dot{\boldsymbol{q}} - T + V = T + V$, where M is the mass matrix of the system. Here the Hamiltonian function H is exactly the total energy of the system. Therefore, Eq. 9 shows that systems described with Hamiltonian mechanics follow the law of conservation of energy.

3.2 Symmetry, invariant and equivariant functions

A symmetry defined on an object or system is some transformation that keeps certain properties unchanged. Typical symmetries include shifts in visual object classification problems, rotations in molecule property prediction problems, and permutations in particle systems. For one object or system, its symmetries form a *symmetry group*, where the following rules are satisfied: (1) associativity, (2) identity, (3) inverse, (4) closure. Some common symmetry groups are: $T(n)$ (n-dim translation group), $O(n)$ (n-dim distance-perserving group including both rotation and reflection), $SO(n)$ (n-dim rotation group), $E(n)$ (n-dim Euclidean group, including translation, rotation and reflection).

On a set Ω representing a domain, we can define a *group action* as a mapping $(\mathfrak{g}, u) \mapsto \mathfrak{g}.u$, where $\mathfrak{g}$ is one element of a symmetry group $\mathfrak{G}$, and $u, \mathfrak{g}.u \in \Omega$. A group action shall satisfies associativity, i.e., $\mathfrak{g}.(\mathfrak{h}.u) = (\mathfrak{g}\mathfrak{h}).u$ for all $\mathfrak{g}, \mathfrak{h} \in \mathfrak{G}$ and $u \in \Omega$. In comparison, the *group action* on a signal space $\mathcal{X} : \Omega \to \mathbb{R}^n$ is defined as:

$$(\mathfrak{g}.x)(u) = x(\mathfrak{g}^{-1}u). \tag{10}$$

We can verify that the above definition satisfies the associativity:

$$(\mathfrak{g}.(\mathfrak{h}.x))(u) = (\mathfrak{h}.x)(\mathfrak{g}^{-1}(u)) = x((\mathfrak{g}\mathfrak{h})^{-1}u) = ((\mathfrak{g}\mathfrak{h}).x)(u). \tag{11}$$

On a domain $\Omega \in \mathbb{R}^n$, we can define an n-dimensional real *representation* of a group $\mathfrak{G}$ as a map $\rho : \mathfrak{G} \to \mathbb{R}^{n \times n}$, connecting each $\mathfrak{g} \in \mathfrak{G}$ to an invertible matrix $\rho(\mathfrak{g})$, and satisfying the associtivity $\rho(\mathfrak{g}\mathfrak{h}) = \rho(\mathfrak{g})\rho(\mathfrak{h})$ for all $\mathfrak{g}, \mathfrak{h} \in \mathfrak{G}$.

For a symmetry group $\mathfrak{G}$ we have the definitions of $\mathfrak{G}$-invariant and $\mathfrak{G}$-equivariant. A function $f : \mathcal{X}(\Omega) \to \mathcal{Y}$ is $\mathfrak{G}$-invariant if $f(\rho(\mathfrak{g})x) = f(x)$ for all $\mathfrak{g} \in \mathfrak{G}$ and $x \in \mathcal{X}(\Omega)$, and it is $\mathfrak{G}$-equivariant if $f(\rho(\mathfrak{g})x) = \rho(\mathfrak{g})f(x)$.

By stacking multiple neural network layers (functions), each of which satisfies either equivariance or invariance under some symmetry group, we can incorporate the knowledge of symmetries of domains into the resulting network. Table 1 [146] connects some neural network architectures with their corresponding symmetry groups and domains.

3.3 Numerical methods for Partial Differential Equations (PDEs)

A generic form of PDEs describing the evolution of a continuous value $v(x, t)$ is as follows:

$$\frac{\partial v}{\partial t} = F(t, x, v, \frac{\partial v}{\partial x_i}, \frac{\partial^2 v}{\partial x_i \partial x_j}, \ldots). \tag{12}$$

Finite Difference Method

Finite difference methods approximate spatial derivatives on the right hand side of Eq. 12 at a certain point x_0 as a linear combination of function values at N neighbors of x_0. In the 1-D case, the approximation of the l-th order spatial derivative can be formulated as follows (here we omit t):

$$\left.\frac{\partial^l f}{\partial x^l}\right|_{x=x_0} = \sum_{n=1}^{N} \alpha_n f(x_0 + h_n). \tag{13}$$

$\alpha_1, \alpha_2, \ldots, \alpha_n$ can be solved via expanding $\alpha_1 f(x_0 + h_1), \alpha_2 f(x_0 + h_2), \ldots, \alpha_N f(x_0 + h_N)$ at x_0 to the N-th order:

$$\begin{cases} \alpha_1 f(x_0 + h_1) & = \alpha_1 \sum_{k=0}^{N-1} \frac{1}{k!} \frac{\partial^k f}{\partial x^k} h_1^k + O(h_1^{N-1}) \\ \quad \cdots & \\ \alpha_N f(x_0 + h_N) & = \alpha_N \sum_{k=0}^{N-1} \frac{1}{k!} \frac{\partial^k f}{\partial x^k} h_N^k + O(h_N^{N-1}) \end{cases}, \tag{14}$$

Table 1 Neural network architectures, symmetry groups and domains

Architecture	Symmetry group $\mathfrak{G}$	Domain Ω
CNN	Grid	Translation
Spherical CNN [147]	Sphere / SO(3)	Rotation SO(3)
Intrinsic Mesh CNN [137, 138]	Manifold	Isometry Iso(Ω) / Gauge symmetry SO(2)
GNN	Graph	Permutation Σ_n
Deep Sets [148]	Set	Permutation Σ_n
Transformer	Complete Graph	Permutation Σ_n
LSTM	1D Grid	Time warping

which can be summed together as

$$\sum_{n=1}^{N} \alpha_n f(x_0+h_n) = \sum_{k=0}^{N-1} \left(\frac{1}{k!} \sum_{i=1}^{N} \alpha_i h_i^k\right) \frac{\partial^k f}{\partial x^k} + O(\max_{i\in[N]} h_i^{N-1}). \tag{15}$$

To solve $\alpha_1, \alpha_2, \ldots, \alpha_n$ for a certain order l, we can let the multiplier of $\frac{\partial^k f}{\partial x^k}$ be 1 and others be 0, and solve the following linear equations:

$$\begin{bmatrix} 1 & 1 & \ldots & 1 \\ h_1^1 & h_2^1 & \ldots & h_N^1 \\ \vdots & \vdots & & \vdots \\ h_1^l & h_2^l & \ldots & h_N^l \\ \vdots & \vdots & & \vdots \\ h_1^{N-1} & h_2^{N-1} & \ldots & h_N^{N-1} \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_N \end{bmatrix} = \begin{bmatrix} 0! \cdot 0 \\ 1! \cdot 0 \\ \vdots \\ l! \cdot 1 \\ \vdots \\ (N-1)! \cdot 0 \end{bmatrix}, \tag{16}$$

the solution $\alpha_1^*, \ldots, \alpha_N^*$ of which satisfies

$$\frac{\partial^k f}{\partial x^k} = \sum_{n=1}^{N} \alpha_n^* f(x_0 + h_n) + O(\max_{i\in[N]} h_i^{N-1}). \tag{17}$$

The above method can be extended to multi-dimensional cases with multi-dimensional Taylor expansion. After approximating spatial derivatives on the left hand side, Eq. 12 can be solved with standard techniques of numerical integration. While Eq. 16 can fully determine all coefficients, recent works [149, 150] relax this by removing some constraining equations and use neural networks to learn undetermined coefficients to combine prior knowledge with stronger expressivity.

Finite Volume Method Unlike the finite difference method, the finite volume method represents the value $v(x, t)$ with its averages over a grid cell: $v_i(t) = \Delta x^{-1} \int_{x_i-\Delta x/2}^{x_i+\Delta x/2} v(x', t)dx'$. Similar to the finite difference method, spatial derivatives can also be estimated as the linear combination of averaged cell values: $\frac{\partial^l v}{\partial x^l} = \sum_{i=1}^{N} \alpha_i^{(l)} v_i$, and the coefficients are estimated in the same way as finite difference method.

For finite volume methods, the equation must be able to be rewritten as:

$$\frac{\partial v}{\partial t} = \frac{\partial}{\partial x} J(t, x, v, \frac{\partial v}{\partial x_i}, \frac{\partial^2 v}{\partial x_i \partial x_j}, \ldots), \tag{18}$$

where J is called a *flux* and has an analytical form derived from the original PDE. The evolution along time can be carried out in following steps: (1) spatial derivatives are estimated on the boundary between grid cells; (2) the flux J is calculated with approximated derivatives using its analytical form; (3) the temporal derivative of averaged cell values is calculated via subtracting J at the cell's left and right boundaries. The final step can be conducted with techniques that promote stability, such as monotone numerical fluxes and Godunov flux [44]. Recent works have integrated data-driven models in step (1) to improve the estimation of spatial derivatives. For example, [44] estimate coefficients with the combination of neural network results and numerical results, and [151] uses policies trained via reinforcement learning to determine coefficients for estimation.

Finite Element Method Finite element methods divide a space into small parts (elements) and approximate the PDE on each. As illustrated in [152], we use the Poisson equation to show the basic idea of the finite element method.

The formulation of the Poisson equation is as follows:

$$\begin{aligned} -\Delta u &= \lambda \quad \text{in } \Omega \\ u &= 0 \quad \text{on } \Gamma. \end{aligned} \tag{19}$$

In finite element method, we multiply a *test function* v on both sides, integrate over Ω, and use integration in parts to

derive the *weak formulation*:

$$a(u, v) = l(\lambda, v), \text{ for all } v \in \mathbb{V}, \tag{20}$$

$$\text{where } a(u, v) = \int_\Omega \nabla u \cdot \nabla v dx, \; l(\lambda, v) = \int_\Omega \lambda v dx. \tag{21}$$

Then we construct a finite-dimensional subspace $\mathbb{V}_h \subset \mathbb{V}$, where $\mathbb{V}_h$ is a piece-wise polynomial function space spanned via $\{\phi_1, \phi_2, \ldots, \phi_n\}$. To solve the weak formulation, we transform it to the *Galerkin weak formulation*, which is an approximation of Eq. 20:

$$\boldsymbol{A}\boldsymbol{u} = \boldsymbol{f}, \tag{22}$$

where $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, $A_{ij} = a(\phi_i, \phi_j)$, $\boldsymbol{u} \in \mathbb{R}^n$ is the solution vector and $\boldsymbol{f} \in \mathbb{R}^n$ is the source vector with $f_i = l(\phi_i, \lambda)$. The solution of Eq. 22 gives the best solution of the PDE in $\mathbb{V}_h$.

3.4 Koopman theory

Given a non-linear dynamic system with its state vector at time t denoted as $\boldsymbol{s}_t \in \mathbb{R}^m$, we can represent the system as $\boldsymbol{s}_{t+1} = F(\boldsymbol{s}_t)$. As defined in [57], the Koopman operator $\mathcal{K}_F$ is a linear transformation defined on a function space $\mathcal{F}$ by $\mathcal{K}_F g = g \circ F$ for every $g : \mathbb{R}^m \to \mathbb{R}$ that belongs to the infinite-dimensional Hilbert space $\mathcal{F}$. With the definition we have $\mathcal{K}_F g(\boldsymbol{s}_t) = g \circ F(\boldsymbol{s}_t) = g(\boldsymbol{s}_{t+1})$.

The Koopman theory [57] guarantees the existence of $\mathcal{K}$, but in practice we often assume the existence of an invariant finite-dimensional subspace $\mathcal{G}$ of $\mathcal{F}$ spanned by k bases $\{g_1, g_2, \ldots, g_k\}$. Define $\boldsymbol{g}_t = [g_1(\boldsymbol{s}_t), g_2(\boldsymbol{s}_t), \ldots, g_k(\boldsymbol{s}_t)]^T$ and $\boldsymbol{g}_{t+1} = [g_1(\boldsymbol{s}_{t+1}), g_2(\boldsymbol{s}_{t+1}), \ldots, g_k(\boldsymbol{s}_{t+1})]^T$, under the assumption we have $\boldsymbol{g}_t, \boldsymbol{g}_{t+1} \in \mathcal{G}$ and there exists a Koopman matrix $\boldsymbol{K} \in \mathbb{R}^{k\times k}$ s.t. $\boldsymbol{g}_{t+1} = \boldsymbol{K}\boldsymbol{g}_t$. The key problem is to find the pair of mappings between the state space $\mathbb{R}^m$ and the invariant subspace $\mathcal{G} \in \mathbb{R}^k$: $\boldsymbol{g} : \mathbb{R}^m \to \mathbb{R}^k$ and $\boldsymbol{g}^{-1} : \mathbb{R}^k \to \mathbb{R}^m$. Recent works [61–63] utilize neural networks as $\boldsymbol{g}$ and $\boldsymbol{g}^{-1}$ to find the mappings in a data-driven way.

4 Methods of PIML

Typical solutions to a problem with machine learning involve three key parts: data, model, and optimization, each of which can be integrated with prior physics knowledge. In the following parts, we will introduce existing techniques of incorporating physics knowledge to each part respectively. It's worth noting that these techniques are not mutually exclusive: physics knowledge can be integrated jointly in more than one of the following ways within the same machine learning-based approach.

In Figure 1, we classify existing works based on forms of physics knowledge and their method of integration. Figure 2 complements this breakdown by visually depicting a generalized PIML pipeline, highlighting how each method of integrating prior physics knowledge plays a role in the approach. We note that for domain knowledge taking analytical forms, existing works integrate the knowledge into all three aspects: data, model, and optimization. However, research works on integrating other general types of physics knowledge, including energy conservation laws, symmetries, numerical methods for PDEs, and Koopman theory, mainly focus on incorporating corresponding knowledge into computation graphs. The main reason is that such general physics knowledge can be used to construct network architectures with inductive biases that more closely match the problem at hand, and this has advantages over data augmentation and physics knowledge-based loss functions in terms of prediction performance and data efficiency [146]. This is due in part to several factors, including (1) general physics knowledge applies to various problems and thus can lead to general network architectures, and (2) has simpler forms that can be translated to combinations of a limited number of differentiable operators, at least when compared to complex numerical simulators designed for domain-specific problems (e.g., weather and turbulence).

4.1 Physics-informed data enhancement

4.1.1 Data generated from simulation

The Universal Approximation Theorem [113] guarantees that multi-layer neural networks with as few as one hidden layer can approximate any continuous function from one finite dimensional space to another arbitrarily well. Therefore, one straight forward method to incorporate physics knowledge into neural networks is to generate training data from the desired physics knowledge. When the size of such data is large, and the neural network is sufficiently expressive and trained properly, the trained model will be able to approximate the behavior of the physics knowledge governing the data generation. Neural network models can also often be accelerated with hardware (e.g., GPUs, FPGAs, ASICs), and can thus act as good surrogate models with much lower computation costs.

[26] present a benchmark dataset from results of numerical global weather simulation with high computational cost, and provide scores of deep learning models. Results show that data-driven models trained with simulation data can achieve results that are competitive with numerical solutions while enjoying greater computational efficiency. [30, 31, 153] utilize different variants of message passing graph

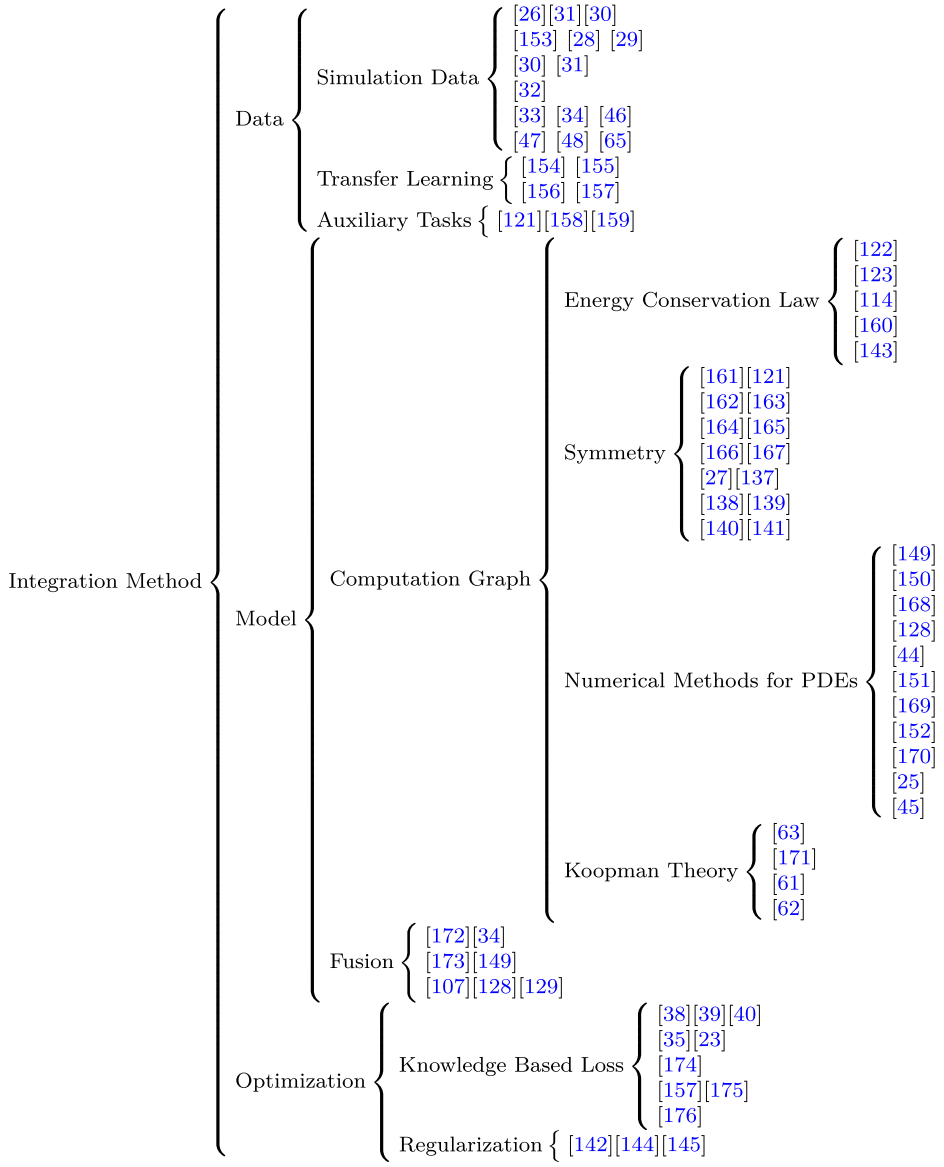

Fig. 1 Existing works classified based on forms of physics knowledge and integration methods. For certain types of knowledge forms, inductive bias integrated in reusable computation graphs have advantages over integration with data and optimization

neural networks [118, 119, 177] and train them with the simulation data of particle systems. Compared to the simulation methods responsible for generating the datasets, the trained surrogate model can accurately predict the dynamics of a wide range of physical systems within the same architecture, and runs orders of magnitude faster.

4.1.2 Transfer learning

For real-world tasks suffering from data limitation or labeling difficulties, the integration of prior physics knowledge about the tasks is critical. Simulators constructed with such physics knowledge can provide large amount of data with high label quality, and can be used to pre-train models. However, the differences between the target real-world data distributions and simulation data distributions call for techniques of transfer learning to mitigate the gap.

[154] present a system for training object detection models with synthetic images. This work adopts the technique of domain randomization, where important parameters of simulators - including lighting, pose, object textures - are randomized in unrealistic ways to encourage the model to learn essential features. [155] uses off-the-shelf simulators to render synthetic data for training a system together with pixel-level domain adaptation between synthetic images and real-world ones. The utilization of synthetic data reduces the required amount of real-world samples by up to 50 times. [156] transfer driving policies trained from simulation

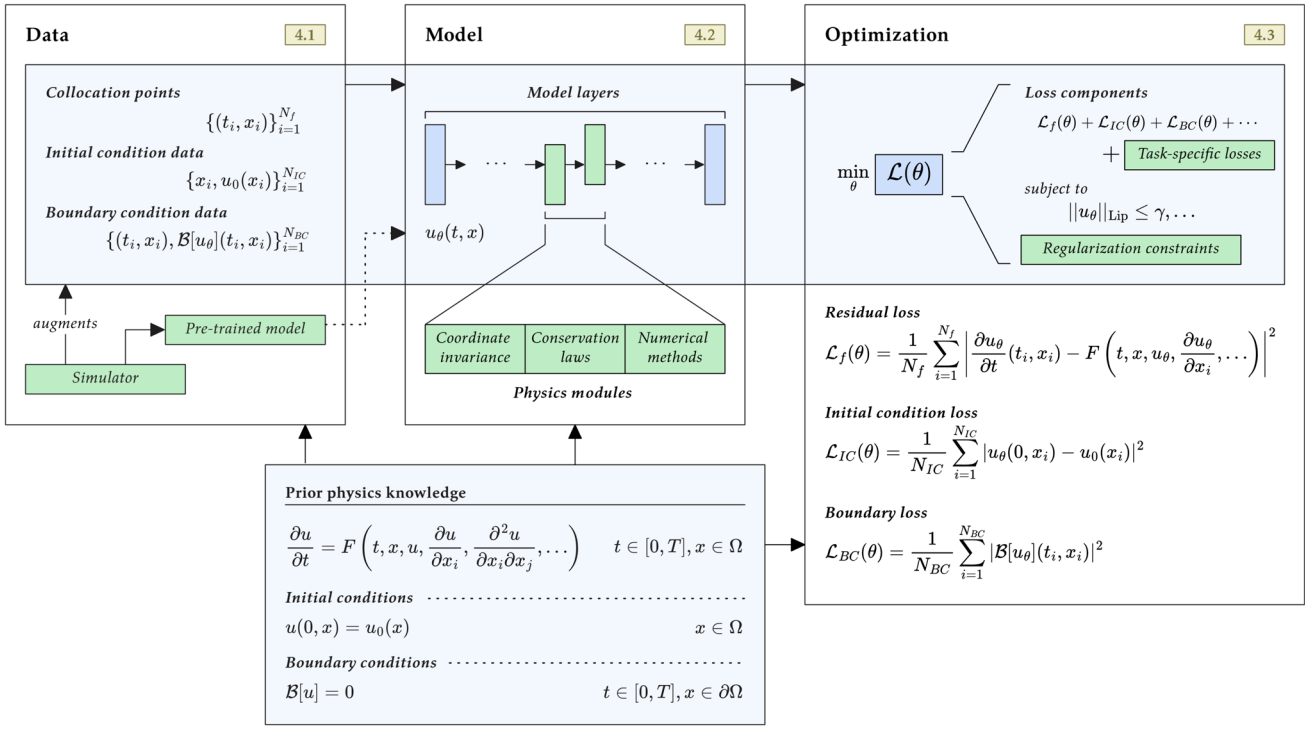

Fig. 2 Depiction of the general PIML elements introduced in Section 4. The three numbered columns coincide with the "integration method" divisions from Fig. 1, namely **data** (Section 4.1), **model** (Section 4.2), and **optimization** (Section 4.3). The blue row stretching across columns highlights the general flow of neural networks with these physics-informed elements (PINNs serving as the primary architectural basis). The accentuated green boxes further show the specific topic breakdowns discussed in each subsection

to reality via modularity and abstraction, where the driving policy is exposed to segmentation results of input scenes and target way points, instead of raw perceptual input or low-level vehicle dynamics. [157] incorporate the task-specific prior knowledge into the model and pre-train it with synthetic data generated by imperfect physical models, which allows the model to get close enough to the target solution and only a small amount of real-world data is needed for refining.

4.1.3 Multitask learning and meta learning with auxiliary tasks

Synthetic data generated from physics-based simulators can also be used to construct auxiliary learning tasks for improving the model's performance on the target task with techniques of multitask learning and meta learning. [121] constructs the auxiliary task as the prediction of interatomic forces for the main task of molecular energy prediction. The labels of the auxiliary task and the main task are generated from simulation at the same time, while the prediction of the auxiliary task is produced via differentiating the energy prediction model. Both tasks are used to train the model simultaneously. [158] adopts the multitask learning scheme by learning shared representations between multiple related PDEs, which are generated by varying coefficients, for better generalizability of the proposed neural network based PDE solver. [159] proposes a spatiotemporal forecasting model with decoupled spatial and temporal modules, where the spatial module is PDE-independent and are trained via model agnostic meta learning (MAML) [178] for fast adaptation on new tasks, while the task-dependent temporal module is trained from scratch for each task.

4.2 Physics-informed neural network architecture design

4.2.1 Physics-informed computation graph

A common means of constructing physics-informed neural network architectures is designing computation graphs that mimic the behavior of the underlying physics knowledge-based methods. While the specific method is highly dependent on the physics in question, the general idea is to start from some existing physics based solution and replace difficult-to-estimate variables with outputs of learnable models (like neural networks), or relax some fixed parameters by enabling them to adapt to the data. In this section, we

introduce several knowledge-specific neural network architectures that directly fuse deep learning models with physics based solutions into a hybrid model.

Energy Conservation Laws

As introduced in Section 3.1, Lagrangian and Hamiltonian mechanics are robust frameworks that enforce energy conservation laws. A series of recent works develop neural network architectures based on these laws to fundamentally incorporate the notion of energy conservation. [122] proposes a network topology titled Deep Lagrangian Networks (DeLaN) encoding the Lagrange-Euler PDE originating from Lagrangian Mechanics, which can be trained with standard optimizers while maintaining physical plausibility. [123] designs Lagrangian Neural Networks (LNNs) to model arbitrary Lagrangian functions via neural networks, and solve the dynamics of the system with a numerical expression derived from the Euler-Lagrangian equation, where gradients from auto differentiation are utilized. Similarly, [114] (HNN) models the Hamiltonian function with a neural network. The derivatives of spatial coordinates and momentum with respect to time are derived from Eq. 8. [160] proposes Neural Hamiltonian Flows (NHF), which are powerful normalizing flow models using Hamiltonian dynamics as the invertible function to model expressive densities. In NHF, the density is decomposed into the "coordinate" part and the "momentum" part in the hidden space, both of which are then propagated with Eq. 8. The propagation is (1) invertible and (2) has the volume ("energy" with respect to the hidden space) preserving property, which satisfies the requirement of normalising flows. Compared to other flow-based approaches, NHF enjoys higher computational efficiency since it avoids the expensive step of calculating the trace of Jacobians. [143] designs the computation graph of the proposed neural network following the Hamiltonian dynamics with control to incorporate the corresponding inductive bias. It further proposes a parameterization that can enforce the Hamiltonian mechanics with coordinates embedded in a high-dimensional space or velocity data instead of momentum.

Symmetry

Table 1 in Sec 3.2 demonstrated the connection between several widely used neural network architectures and their representative symmetry groups. Here we introduce methods that incorporate other types of symmetry groups into neural network architectures. Based on the representations of the adopted symmetry group, introduced methods can be categorized by using (1) invariant treatment of coordinates, (2) irreducible representations, or (3) regular representations.

Invariant Treatment of Coordinates Depending on the symmetry group, spatial coordinates should be properly processed instead of being used as raw inputs. [161] develops equivariant message passing to E(3) by ensuring messages passed among nodes only depend on distances, which follows the property that E(3) preserves distances between nodes. The same technique is also used in [121]. [162] propose a set of transformation invariant and equivariant GNN models by tweaking the definition *isometric adjacency matrices*, which can be viewed as a weighted adjacency matrix for each direction and reflects spatial information.

Irreducible Representations All elements of a roto-translation group can be transformed into an irreducible form: a vector that is rotated by a block diagonal matrix. The full set of equivariant mappings for some symmetry group can be solved with equivariance constraints over convolution kernels. The solutions form a linear combination of equivariant basis matrices, which can be used for equivariant convolutions. [163] gives a general solution of the kernel space constraint for arbitrary representations of the Euclidean group E(2) and its subgroups, which forms a wide range of equivariant network architectures. [164] develops convolution filters locally equivariant to 3D rotations, translations, and permutations, which are built from spherical harmonics. [165] further enhances [164] with the self-attention mechanism.

Regular Representations Regular representation approaches store copies of latent feature embeddings for all elements of a symmetry group. To mitigate this computational burden, some recent works such as [166, 167] use *Lie groups* as the tool for rapid prototyping across various symmetry groups. Only the exponential and logarithm maps are required for incorporating equivariance to a new symmetry group.

Numerical Methods

Section 3.3 introduces several numerical methods for solving PDEs. In this paragraph, we introduce some recent works integrating numerical solutions for each method.

Finite Difference Method [149, 150] propose learnable differential operators by learning convolution kernels to approximate unknown nonlinear responses in PDEs. All kernels are properly constrained by fully exploiting the relation between the orders of differential operators and the orders of sum rules of filters, which originates from wavelet theory. These constraints ensure both the model's ability to identify PDEs and its expressivity. [168] proposes an efficient convolution kernel on unstructured grids of spherical signals using parameterized finite difference operators. [128] leverages differences of sparsely available data from physical systems via the spatial difference layer (SDL). SDL is inspired by finite difference operators on graph and triangulated mesh and replaces fixed parameters in these operators with the output of a GNN capturing spatial information of the input data.

Finite Volume Method [44] adopts CNNs to generate coefficients for approximating spatial derivatives, followed by the standard finite volume method. CNNs are optimized end-to-end to best satisfy the equations on low resolution grids, and produce accurate numerical results: it can be integrated at resolutions 4-8x coarser than is possible with standard finite volume methods. [151] creates new PDE solvers based on the WENO scheme [179] via generating its coefficients from a learned policy network trained with reinforcement learning.

Finite Element Method [169] mimics the behavior of finite element analysis: it assigns nodes of a GNN to selected spatial locations and uses message passing on the graph to model the relationship between an initial function and a resulting function defined in the same space. Both the locations of nodes and their connectivity can be optimized to focus on the most important parts of the space. [152] proposes a two-stage optimization framework for PDE-constrained optimization problem. At the first stage, the framework obtains a surrogate model to prediction solutions of finite element method directly from control parameters. At the second stage, the framework performs gradient-based PDE-constrained optimization. [170] introduces convolution operators on unstructured point clouds based on Generalized Moving Least Squares (GMLS), which is a non-parametric technique in finite element method [180] for estimating linear bounded functionals from scattered data.

Koopman Theory

Lusch et al. [63] first combines deep learning models with the Koopman operator. It utilizes the power of deep learning to identify nonlinear coordinates on which the dynamics are globally linear using the Koopman operator. The resulting method benefits from both the power and generality of deep learning models and the physical interpretability of Koopman embeddings. [171] proposes minimization of the residual sum of squares of linear least-squares regression to estimate the encoders and decoders that maps data into the Koopman invariant subspaces where the linear regression fits well. [61] extends deep learning based Koopman operators to scenarios with multiple objects. It adopts GNNs to encode object-centric states and uses a block-wise linear state transition matrix (Koopman matrix) to enforce the shared structure among objects. [62] incorporates the consistency by penalizing the consistency mismatch of forward and backward Koopman matrices.

4.2.2 Fusion of deep learning and physics-based modules

In addition to previously introduced methods that mimic the behaviour of physics based solutions in the computation graph design of neural network layers, deep learning and physics-based methods can also be fused in a higher level: modules constructed with standard neural network blocks and physics rules work interactively but only expose input/output to each other.

Garcia Satorras et al. [172] adopts a graphical model derived from equations of motion in a physics model to predict the next future state while adding it with the predicted residual part from a GNN module. [173] presents HybridNet, a framework combining data-driven deep learning and model-driven computation for reliable spatiotemporal evolution prediction. The deep learning part, Convolutional LSTM (ConvLSTM), works as the backbone to predict the evolution of external input to the system. The model-driven part, Cellular Neural Network (CeNN), transforms numerical computation in PDE solvers and is able to infer unknown physical parameters. [34] proposes a hybrid approach for fluid flow prediction containing two components: one GNN-based module operating directly on the original fine-grained non-uniform mesh used in CFD, and one CFD solver operating on a much coarser resolution. The output of the CFD solver is upsampled to the fine-grained mesh and then concatenated with the hidden embeddings from GNN layers. The hybrid model generalizes better than pure GNN-based approaches and is still faster than directly running CFD simulation on the original mesh. [107, 149] uses constrained convolution kernels to extract approximated spatial derivatives as features, which are fed as the input to the following neural network layers that capture unknown dynamics and produce the final prediction.

4.3 Physics-informed optimization

Prior physics knowledge can also be integrated into the optimization process in the form of loss functions directly derived from task-specific knowledge or regularizations from physics principles. The integration of physics knowledge in optimization targets reshapes the optimization space and encourages the training process to converge to physically plausible solutions.

4.3.1 Task-specific knowledge based loss terms

A variety of physics knowledge can be described in the form of PDEs, which provides connections between the spatial and temporal derivatives as well as constraints of values of interest on boundaries. [35, 38–40] adopts multi-layer perceptrons to directly model the mapping from input spatial and temporal coordinates to the value of PDE solutions. Instead of only minimizing the prediction error between the output and solutions from numerical methods, the series of works adds loss terms enforcing the equation structure, which penalizes the violation of PDEs using derivatives from auto-differentiation

of the neural network solution. [23] optimizes the model for super-resolution of turbulent flows by minimizing a weighted combination of two losses: one is the norm of the difference between predictions and ground truth values, the other is the norm of residuals of the governing PDEs. [174] predicts the traffic flow with a neural network taking time and coordinates as input, and constructs physical discrepancy loss terms with an existing second-order traffic model. [157] trains the proposed lake temperature prediction model with generalized loss function to include the physical consistency-based penalty, which encourages the consistency between lake energy and energy fluxes.

4.3.2 Regularization

[142] adopts a deep neural network to predict the unknown disturbance forces in the controller of drones. To guarantee the system's stability, the authors first derive the overall stability and robustness requirement indicating constraints on the Lipschitz constant, then minimizes its upper bound - the spectral norm of weights in each layer - together with the prediction error in the optimization. [144, 145] both augment purely physics laws/rule-based prediction models with learnable control signals to mitigate the approximation errors of prior knowledge. While they differ in terms of processes for solving PDEs, both have constraints minimizing the norm of control signals, which originates from the least action principle [181].

5 Challenges and future directions

5.1 Challenge 1: Handcrafted Selection of Physics Knowledge for Incorporation

Existing works require expertise of the domain-specific knowledge of tasks to incorporate the most appropriate physics knowledge. While this serves the purpose of leveraging domain-knowledge to mitigate the deficiencies of pure data-driven methods, it lacks the flexibility of identifying the correct physics knowledge depending on the task. For example, [23] directly uses the governing equation of the Rayleigh-Benard instability problem as the prior knowledge for turbulence super-resolution, while [24] chooses the derived Hybrid RANS(Reynolds-averaged Navier-Stokes)-LES(Large Eddy Simulation) Coupling method for turbulence prediction. Although both super-resolution and prediction tasks are defined on the same turbulence system governed by the same physics laws, choosing the form of physics laws (the original form/the derived approximation form) to incorporate is heuristic.

Research Direction 1: Automatic Identification of Proper Physics Knowledge To Incorporate
A particularly promising research objective is reaching an appropriate middle ground between domain-specific knowledge and pure data-driven modeling. Here we discuss some potential approaches to realizing such an optimal trade-off, insofar as it exists.

Neural Architecture Search (NAS) NAS methods facilitate automatic design of neural network architectures and have been shown to outperform manually designed architectures on many tasks, including image classification and semantic segmentation [182]. By restricting the available physics knowledge within a pre-selected search space and enabling the model to discover the optimal knowledge using NAS techniques, the resulting architecture can reach an ideal balance between exploiting prior knowledge and adapting to observed data. [183] presents a neural block dynamics design space of neural network components that encompasses various state-space models as the search space for NAS, and models given by NAS in such a space achieve high accuracy with physically consistent results.

Automatic Modularization of Network Architectures Modularization of network architectures plays a key role in physics-informed network architecture design. For example, DeLaN [122] and HNN [114] contain modules estimating the Lagrangian/Hamiltonian function of the system. [44, 128] attempt to separate modules approximating spatial and temporal derivatives in the network. Modularity provides better generalizability [120], and some modules can further be trained in multiple tasks and benefit the overall performance on all tasks [159, 184]. However, existing works still require a certain selection of physics knowledge to guide the modularization. Recent works have started exploring the automatic discovery of functional modules during the training process. For example, [185] divides models into both reusable (task-invariant) and task-specific modules by estimating the variance of parameters across tasks. [186] lets multiple groups of recurrent cells compete with each other so that they are only updated at time steps where they are most relevant. This enables the model to learn modular structures and leads to improved generalization.

5.2 Challenge 2: lack of benchmarks and evaluations of PIML methods

Comprehensive benchmarks have been shown to greatly boost the development of associated research areas. Examples include the ImageNet Large Scale Visual Recognition Challenge(ILSVRC) [187] and Common Objects in Con-

text(COCO) [188] from computer vision, as well as the Workshop on Statistical Machine Translation (WMT) [189] and Stanford Question Answering Dataset (SQuAD) [190] from natural language processing. Unfortunately, due to the complexity and heterogeneity of problem settings, PIML still lacks comprehensive benchmarks for evaluating various methods of knowledge integration, which creates barriers in the development of PIML. First, most problems in PIML come from physics or engineering applications, where acquiring the data and formalizing the task can be challenging for researchers without domain knowledge and experience. Second, existing works such as [34, 44, 62, 114, 122, 128, 149] heavily rely on heterogeneous domain-specific datasets, which greatly increases the difficulty of fairly comparing different PIML methods.

Research Direction 2: Comprehensive Benchmarks for PIML Methods Constructing comprehensive benchmarks for PIML is of great need for boosting attention, development, and adoption by the community. According to the above discussion, ideal benchmarks (1) must provide publicly available and organized datasets, (2) formulate benchmark tasks as typical machine learning tasks, such as classification and regression, and (3) be general enough to accommodate various PIML methods as well as data-driven and pure physics-based methods. Recent development along this direction includes **WeatherBench** [26] and **Open Graph Benchmark (OGB)** [191]. WeatherBench provides processed weather data together with clearly defined tasks and evaluation metrics for medium-range weather forecasting. It also presents performance of purely data-driven models and numerical models as baselines. OGB offers data of protein structures and molecule structures, and construct node/link/graph property prediction tasks, where performance of PIML methods and data-driven baselines can be directly compared.

5.3 Challenge 3: suboptimal existing neural network architectures and optimization methods for PIML

Established theories and empirical conclusions of neural network architectures and optimization methods are mostly developed in areas where neural network methods were first shown to demonstrate advantages, such as computer vision and natural language processing. The same is not automatically true for physics-based tasks, however, at least not without sufficient empirical or theoretical justification. The reason is that PIML methods usually involve explicit use of gradients in forward processes and objective functions, leading to the existence of high-order derivatives in the backward process, which shapes optimization spaces in a significantly different manner from typical deep learning models. For example, [123] observes that regular parameter initialization methods such as Kaiming [192] and Xavier [193] are insufficient, as the unusual optimization objective is highly nonlinear.[194] also empirically demonstrate that the widely used Rectified Linear Unit (ReLU) activation is not effective in the physics-informed neural network (PINN) architecture proposed by [38, 39].

Research Direction 3: Novel Neural Network Designs for PIML
The drastic differences in network architecture and objectives between PIML and conventional deep learning tasks signify the importance of novel neural network designs for PIML from both the architecture and the optimization aspects.

From the architecture perspective, since many PIML methods involve the utilization of gradients from the auto-differentiation of neural networks, designing new architectures/components better preserving gradient information is one promising direction. [195] leverages periodic activation functions for implicit neural representations and demonstrate that they are ideally suited for representing complex natural signals and their spatial/temporal derivatives.

From the optimization perspective, multiple objectives (including both the task objective and the physics-informed constraints) may contradict each other and lead to sub-optimal results under vanilla optimization methods. [196] notices the discrepancy of updating directions between the boundary constraint loss and the approximation loss in PINNs [38, 39], and proposes the dynamic pulling method (DPM) to align their updating directions, which significantly improves the extrapolation performance of PINNs. [197] provide guidance for working around many practical challenges of applying PINNs, highlighting empirically-backed strategies for overcoming training instability. In particular, the authors highlight important pre-processing steps (e.g., non-dimensionalization of PDEs), key architectural considerations (e.g., incorporating random Fourier embeddings), and effective optimization strategies (e.g., loss balancing, temporal causality preservation).

5.4 Challenge 4: general purpose foundation model of PIML

Building a generalized base model for physics-informed machine learning requires combining the laws of physics with machine learning techniques. One of the main challenges in doing so is handling the non-convexity of physical loss functions, which complicates optimization and convergence during training [198]. In addition, when using architectures designed to avoid overfitting, the capacity of the model is often insufficient, hindering the ability to accurately capture complex physical phenomena. Ensuring that

the model adheres to fundamental physical principles (e.g., conservation laws and boundary conditions) is critical but difficult [199], as violating these principles can lead to implausible predictions. Data requirements are another challenge, as obtaining high-quality samples for training is both costly and time-consuming [200]; incorporating the laws of physics reduces data requirements but adds complexity. Hyperparameter tuning is also demanding, requiring the careful balancing of different loss terms to effectively integrate the governing equations [201]. Finally, achieving good generalization across a variety of systems and ensuring long-term prediction accuracy is critical, as PIML models must extrapolate beyond the training data while remaining faithful to the laws of physics [202]. Addressing these challenges requires innovative architectures for embedding physical constraints into neural networks, improved optimization strategies, and leveraging domain knowledge to improve performance and generalization.

Research Direction 4: Foundation models for Partial Differential Equations (PDEs)

Foundation models for partial differential equations are advanced neural network architectures designed to solve a wide range of PDE problems efficiently and effectively. These models leverage large-scale pretraining and innovative architectures to generalize across different types of PDEs, potentially revolutionizing computational science and engineering. The development of foundation models for PDEs is advancing through innovative approaches such as Unified PDE Solvers (UPS) [203], PDEformer-1 [204], Poseidon [205], SciML [206], and PROSE-PDE [207]. These models employ advanced architectures like FNO-transformers and graph transformers to efficiently solve various spatiotemporal and time-dependent PDEs, demonstrating strong generalization and adaptability across different domains.

Building these models involves significant challenges. One major challenge is the data and computational costs, as developing PDE foundation models is resource-intensive, often requiring vast amounts of data for pretraining which can be prohibitive for complex systems [208]. Additionally, the neural network architecture must be specifically designed to handle the diverse characteristics of PDEs, making it difficult to create models that generalize well across different PDE types.

Ensuring that models can generalize across various PDE systems and effectively transfer knowledge to new problems requires sophisticated pretraining and transfer learning strategies. Enhancing models with domain knowledge, such as physics principles, could improve performance but adds complexity to model design. Moreover, data-driven models often face challenges with error accumulation over long-term predictions, necessitating robust solutions for stability. There is also a need for scalable solutions that efficiently manage large-scale problems without compromising accuracy, with techniques like cross-modal adaptation using pre-trained large language models being explored.

Addressing these challenges involves leveraging novel architectures, efficient training methodologies like pretrain-and-fine-tune paradigms, and integrating domain-specific knowledge to develop robust and versatile PDE foundation models.

6 Summary

In this paper, we provide a thorough and comprehensive survey of existing works in the space of physics-informed machine learning. We summarize methods from three aspects: (1) motivations of PIML, (2) physics knowledge in PIML, (3) methods of knowledge integration in PIML. We additionally discuss existing challenges of PIML and elaborate upon associated potential research directions. Our hope is that this survey provides a useful guide for practitioners and researchers alike, and may facilitate a better understanding of both the current advancements and open research opportunities in the space of PIML.

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Declarations

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